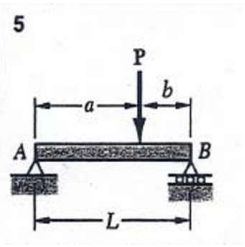
CALCULATION VALIDATION

Case 1: Simply supported beam subjected to point load

Length(L)	=	6	m
	=	6000	mm
Point Load(P)	=	40	kN
	=	40000	N
Load position from left	=	3	m
support(a)	=	3000	mm
Load position from right	=	3	
support(b)	=	3000	
Young's Modulus(E)	=	30000	Mpa
Moment of Inertia(I)	=	8.00E+08	mm4

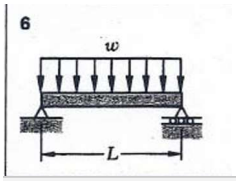


	Calculated	Software
Max shear(kN)	20	20
Max moment(kN-m)	60	60
Max deflection(mm)	7.50E+00	7.5
Max shear location(m)	Just right of support A	Just right of support A
Max moment location(m)	3	3
Max deflection location(m)	3	3

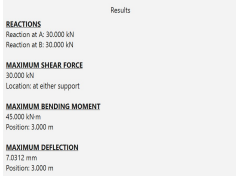


Case 2: Simply supported beam subjected to UDL

Length(L)	=	6	m
	=	6000	mm
UDL(w)	=	10	KN/m
	=	10	N/m
Young's Modulus(E)	=	30000	Mpa
Moment of Inertia(I)	=	8.00E+08	mm4

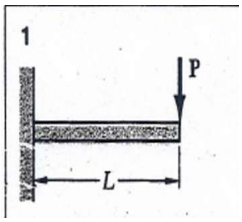


	Calculated	Analysis Software
Max shear(kN)	30	30
Max moment(kN-m)	45	45
Max deflection(mm)	7.03E+00	7.03
Max shear location(m)	Just right of support A	Just right of support A
Max moment location(m)	3	3
Max deflection location(m)	3	3

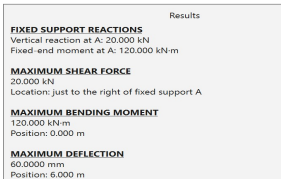


Case 3: Cantilever beam subjected to point load

Length(L)	=	6	m
	=	6000	mm
Point Load(P)	=	20	kN
	=	20000	N
Load position from left	=	6	m
support(a)	=	6000	mm
Load position from free	=	0	
end(b)	=	0	
Young's Modulus(E)	=	30000	Mpa
Moment of Inertia(I)	=	8.00E+08	mm4

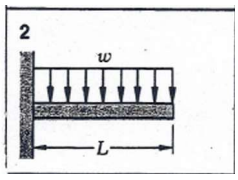


	Calculated	Software
Max shear(kN)	20	20
Max moment(kN-m)	120	120
Max deflection(mm)	6.00E+01	60
Max shear location(m)	Just right of support A	Just right of support A
Max moment location(m)	Just right of support A	Just right of support A
Max deflection location(m)	6	6



Case 4: Cantiliver beam subjected to UDL

Length(L)	=	6	m
	=	6000	mm
UDL(w)	=	10	KN/m
	=	10	N/m
Young's Modulus(E)	=	30000	Mpa
Moment of Inertia(I)	=	8.00E+08	mm4



	Calculated	Analysis Software
Max shear(kN)	60	60
Max moment(kN-m)	180	180
Max deflection(mm)	6.75E+01	67.5
Max shear location(m)	Just right of support A	Just right of support A
Max moment location(m)	Just right of support A	Just right of support A
Max deflection location(m)	6	6

